

TOPICS

- ❖ NETWORK LAYER - BASICS

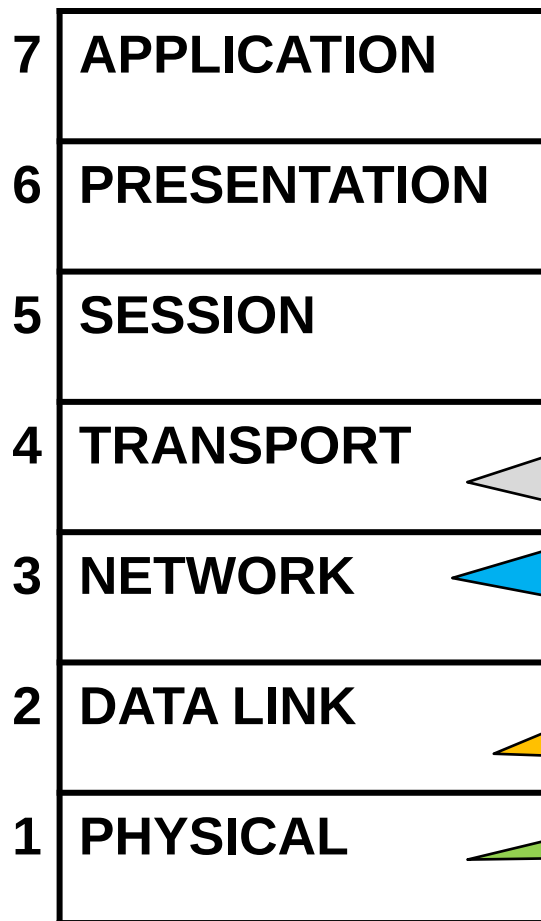
- ❖ NETWORK LAYER - FUNCTIONS

 - ❑ CONNECTION MODEL

 - ❑ ADDRESSING

 - ❑ ROUTING

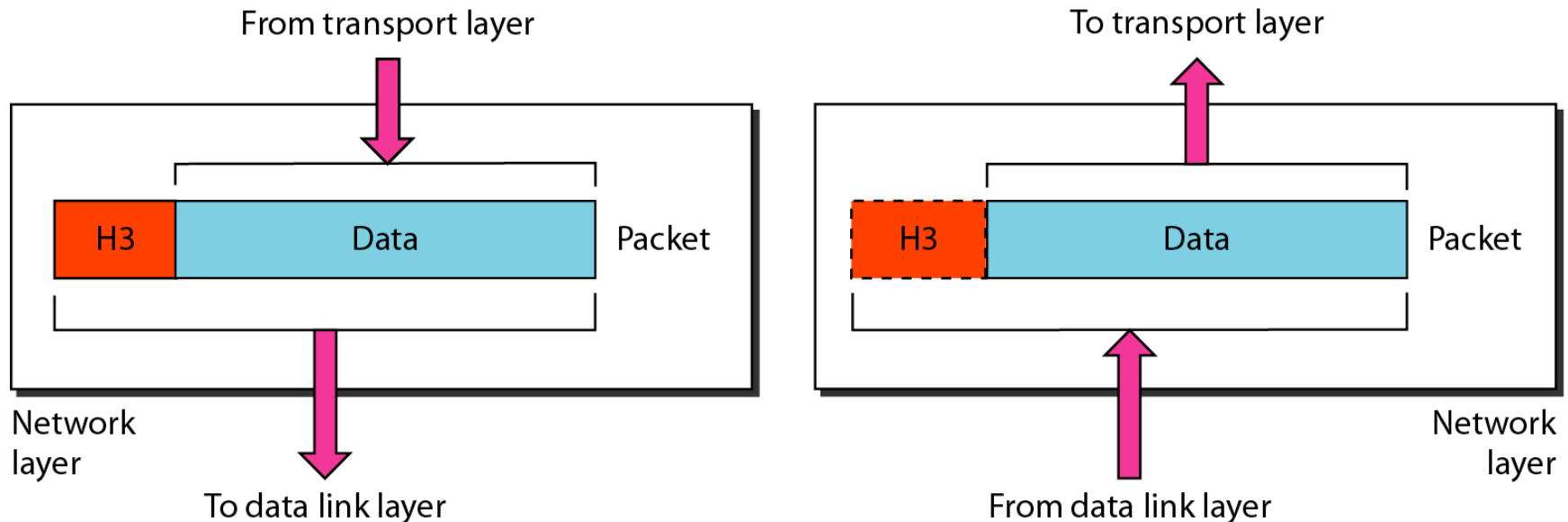
OSI MODEL



- Provide reliable process to process delivery of **Message**, and
- Error recovery

ROLE OF NETWORK LAYER

The network layer is responsible for the delivery of individual **packets** from the **source host** to the **destination host**.



NETWORK LAYER

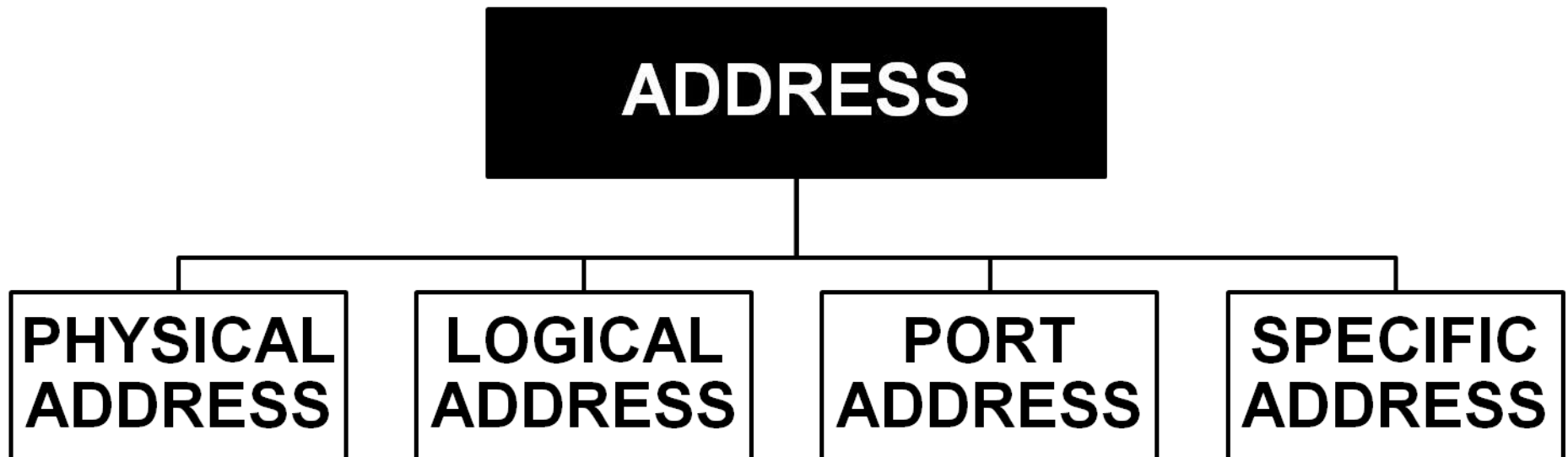
FUNCTIONS

- Host addressing
- Connection model
- Message forwarding

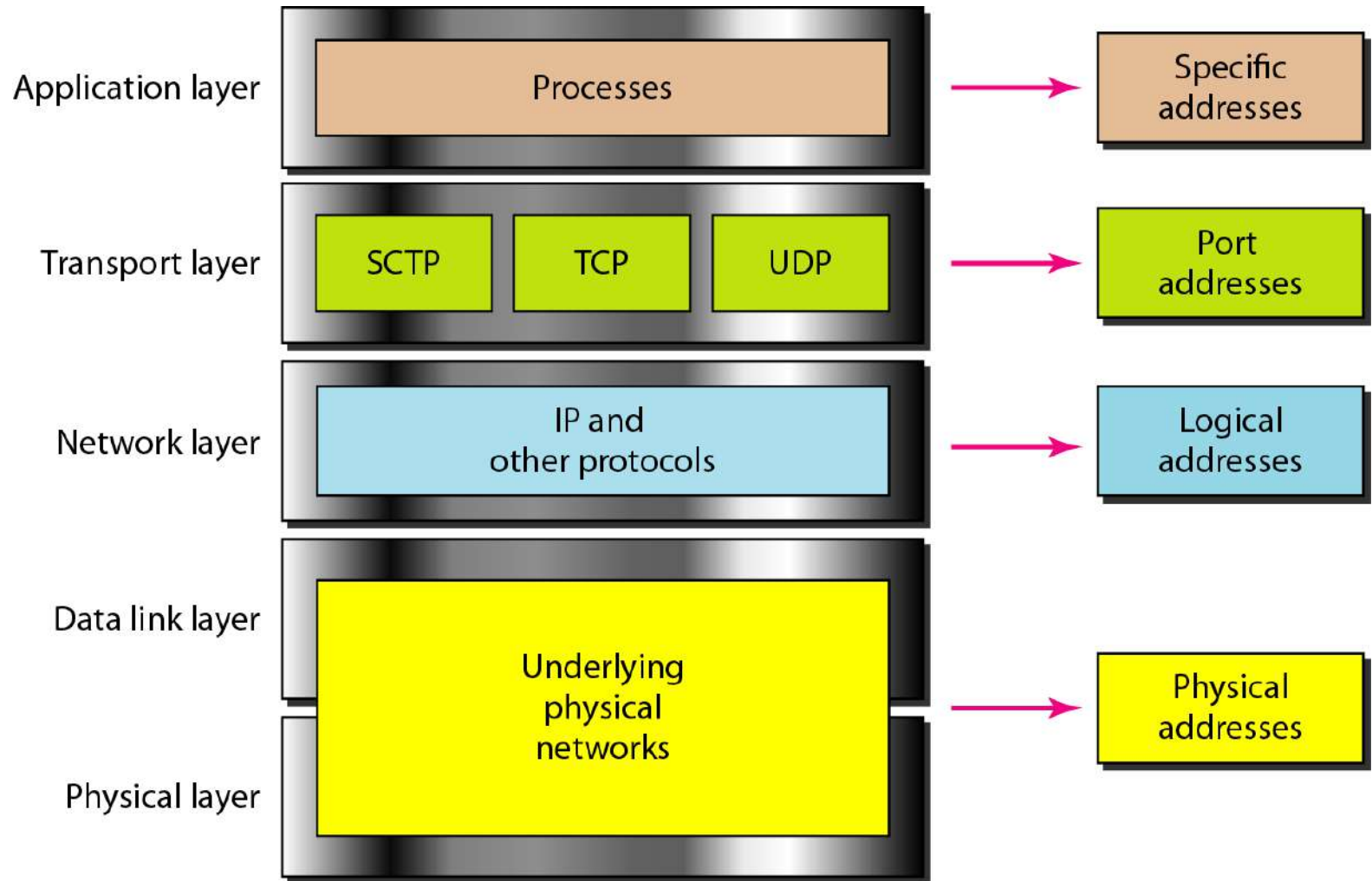
NETWORK ADDRESSES

- Every device on the internet has **address**
- Concept of address helps in distinguishing source, destination and intermediate nodes
- Network entities can easily communicate if they are able to make reference to different addresses

TYPE OF ADDRESSES



LAYER WISE ADDRESSES

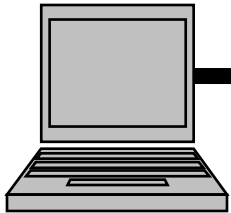


NETWORK LAYER FUNCTIONS

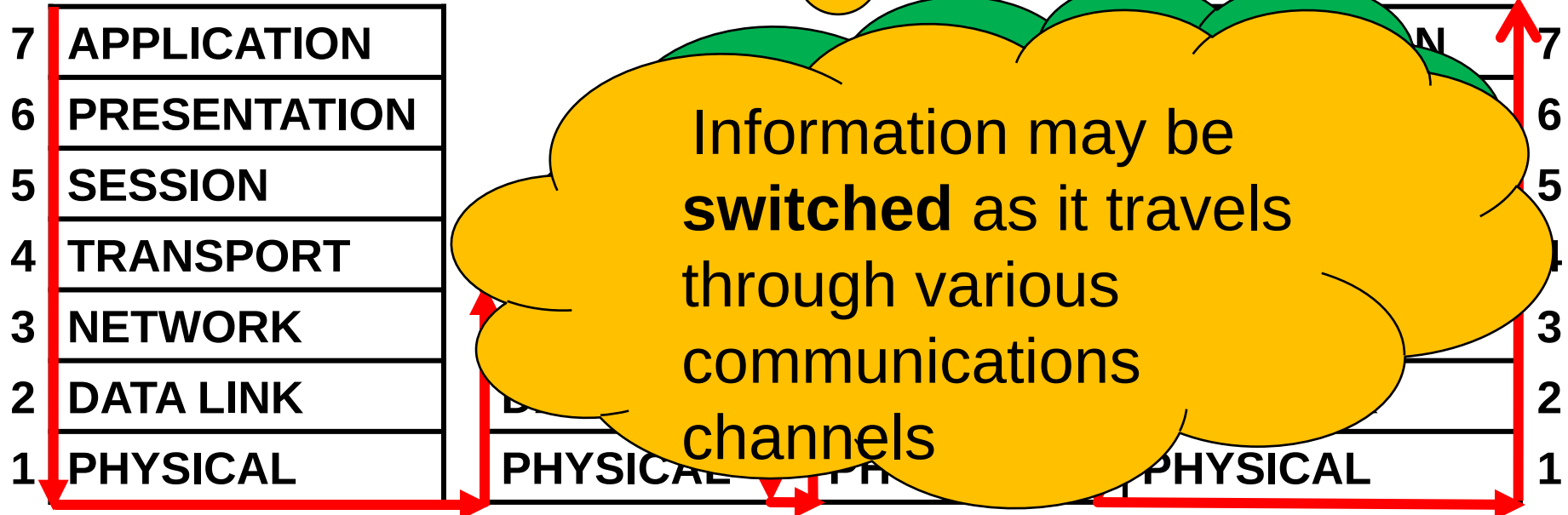
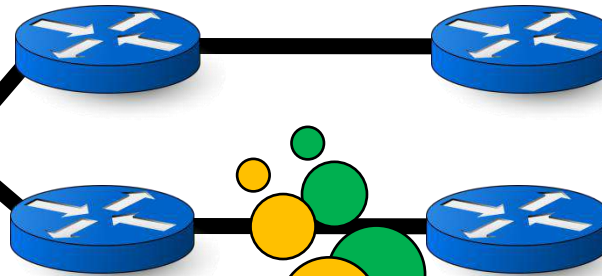
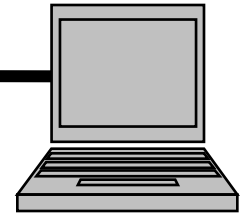
- Host addressing
- **Connection model**
- Message forwarding

OSI MODEL

PC 1



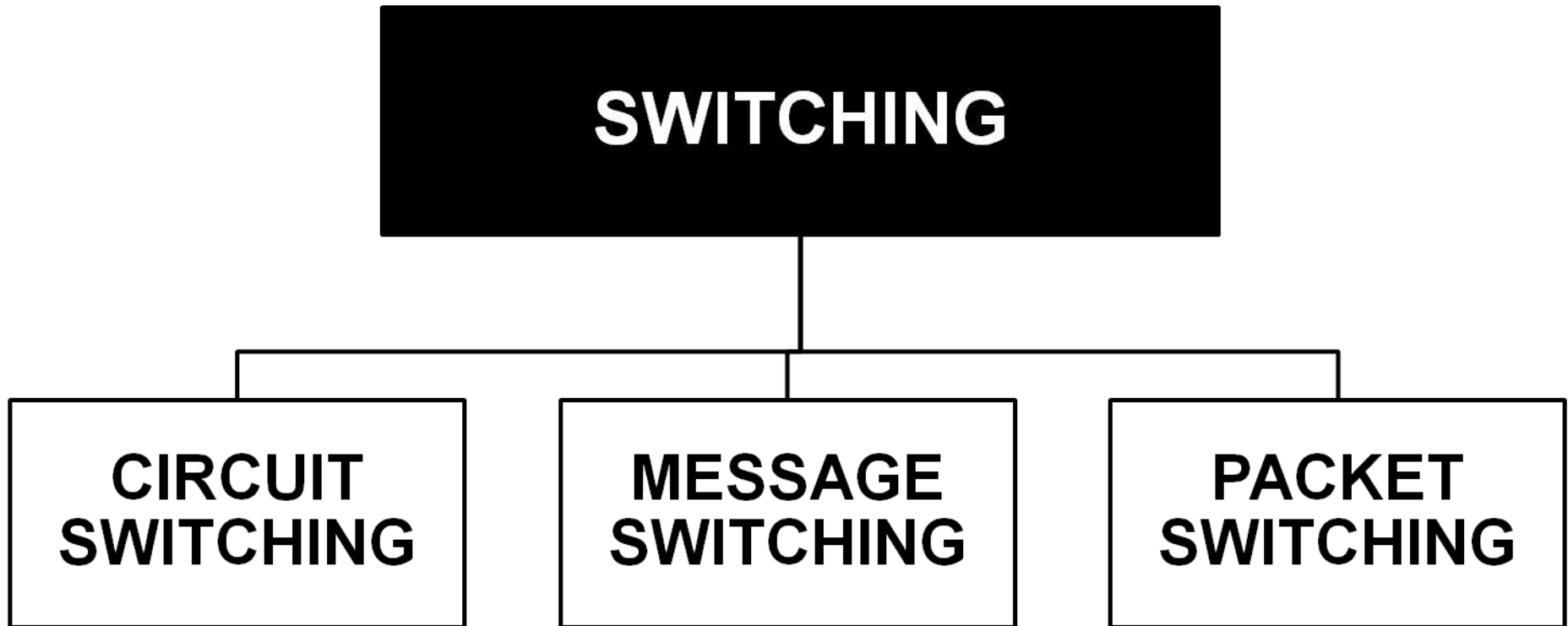
PC 2



SWITCHING

- Switching is a technique that connects the sender and the receiver for communication
- The need arises when you have multiple links between sender and receiver

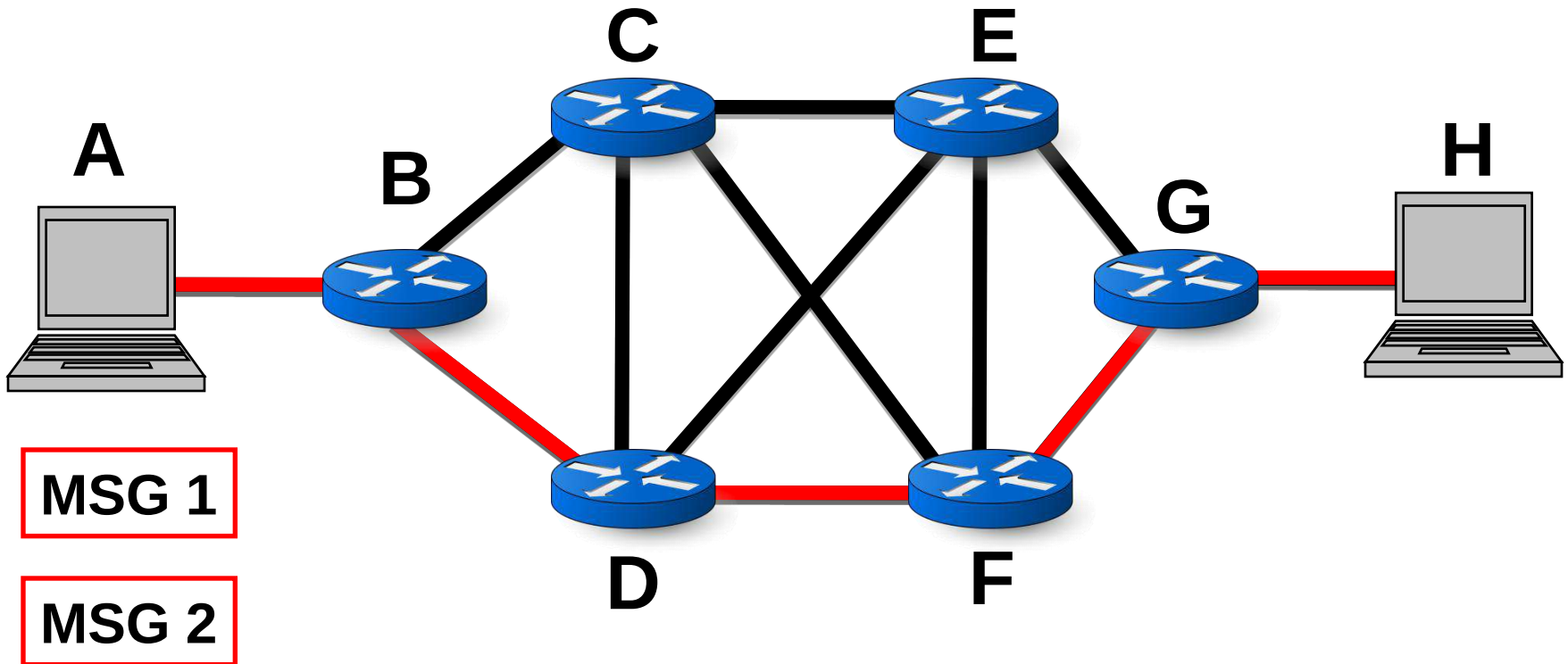
TYPES OF SWITCHING



CIRCUIT SWITCHING

- Circuit switching is a technique that connects the sender and the receiver by a single path for the duration of a conversation
- A complete end-to-end connection must exist before communication can take place

CIRCUIT SWITCHING



CIRCUIT SWITCHING - BENEFITS

- Once a circuit is established, there is virtually **no channel access delay**
- Dedicated transmission channel with **guaranteed data rate**

CIRCUIT SWITCHING - CONSIDERATIONS

- Single, dedicated channel makes **inefficient** use of media
- Dedicated channels are relatively **expensive** in terms of time, money and bandwidth used
- Subjected to **long connection delays**

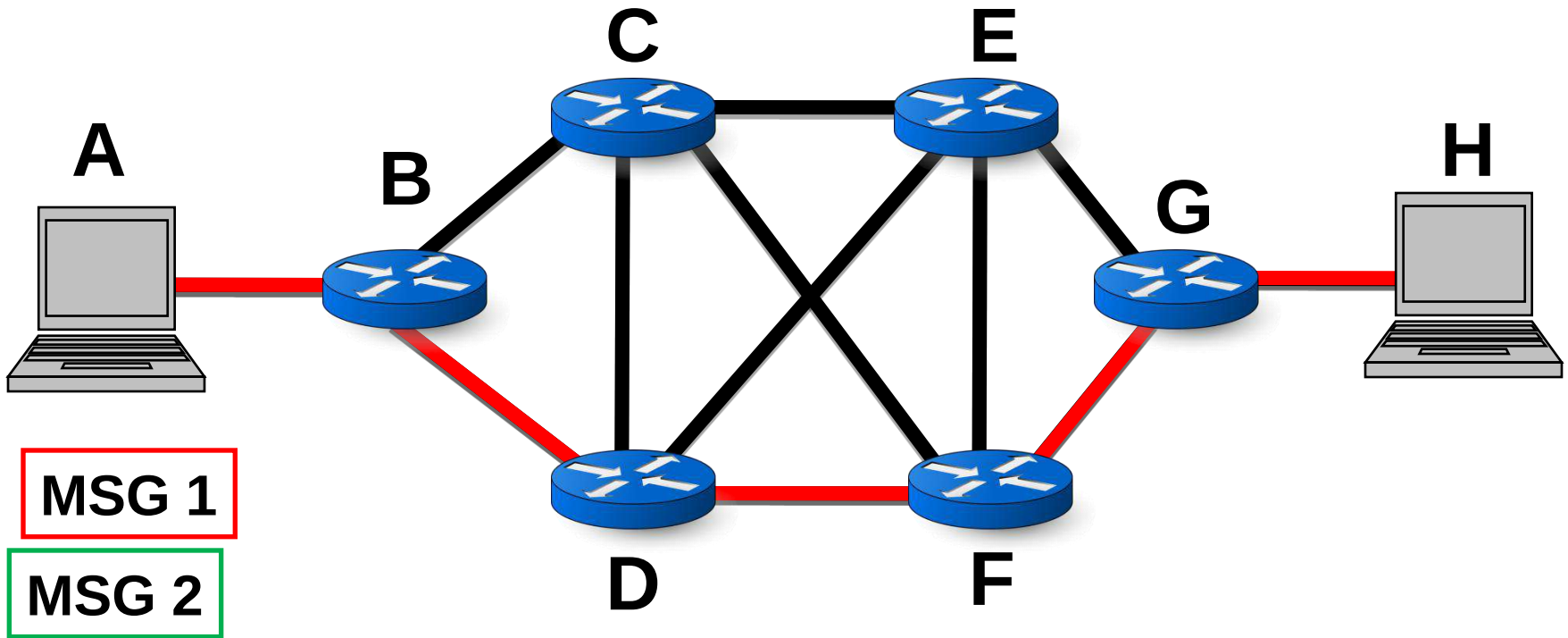
CIRCUIT SWITCHING – SHORTCOMINGS

- **Inefficient** when channel utilized for a small percent of the total time for which connection is maintained
- **Transmission and Receiving rate** of the communicating devices **must be same** which is difficult to achieve for devices supplied by different vendors or having different standards

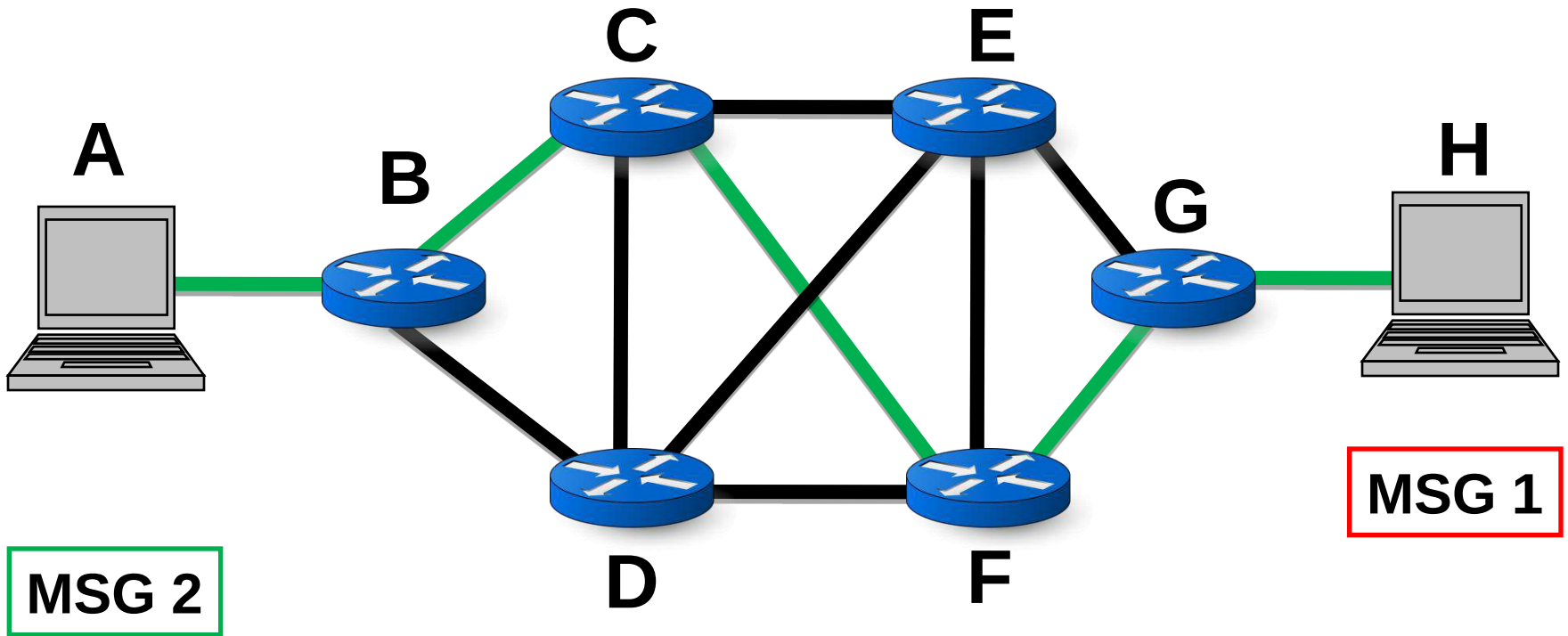
MESSAGE SWITCHING

- No dedicated path between two stations for an entire conversation
- Conversation is divided into messages
- Each message has its destination
- Also known as store-and-forward network
- Support services as E-mail. Workflow, groupware

MESSAGE SWITCHING



MESSAGE SWITCHING



MESSAGE SWITCHING - BENEFITS

- Accommodate more devices by sharing bandwidth
- Reduce network congestion by storing or by prioritizing the messages
- Single message can be sent to multiple devices
- Offline communication possible as in **E-mail**

MESSAGE SWITCHING

- CONSIDERATIONS

- Not suitable, where real time communication required like audio / video communications
- Large and costly storage disks for potentially long messages

PACKET SWITCHING

- Messages are divided into smaller parts, called **PACKETS**
- Each packet is tagged with source, destination and intermediary node addresses
- Packet is small in size; when compared to a **message**, so can be stored in **RAM** instead of hard disks

TYPES OF PACKET SWITCHING

PACKET SWITCHING

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graph TD; A[PACKET SWITCHING] --> B[DATAGRAM PACKET SWITCHING]; A --> C[VIRTUAL CIRCUIT PACKET SWITCHING]
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**DATAGRAM
PACKET
SWITCHING**

**VIRTUAL CIRCUIT
PACKET
SWITCHING**

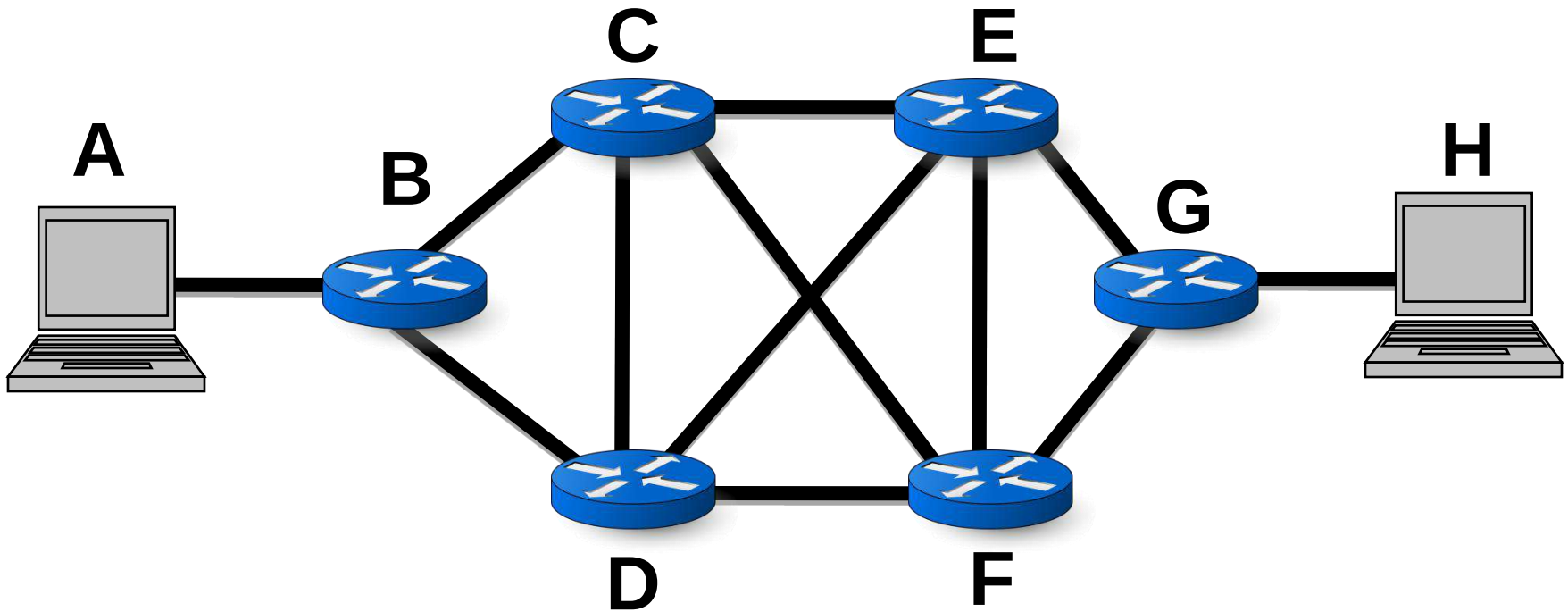
DATAGRAM PACKET SWITCHING

- Very Similar to message switching
- Each message is divided into uniform length information size called **PACKET**
- Each packet is transmitted independently and carries full source and destination address

DATAGRAM PACKET **SWITCHING**

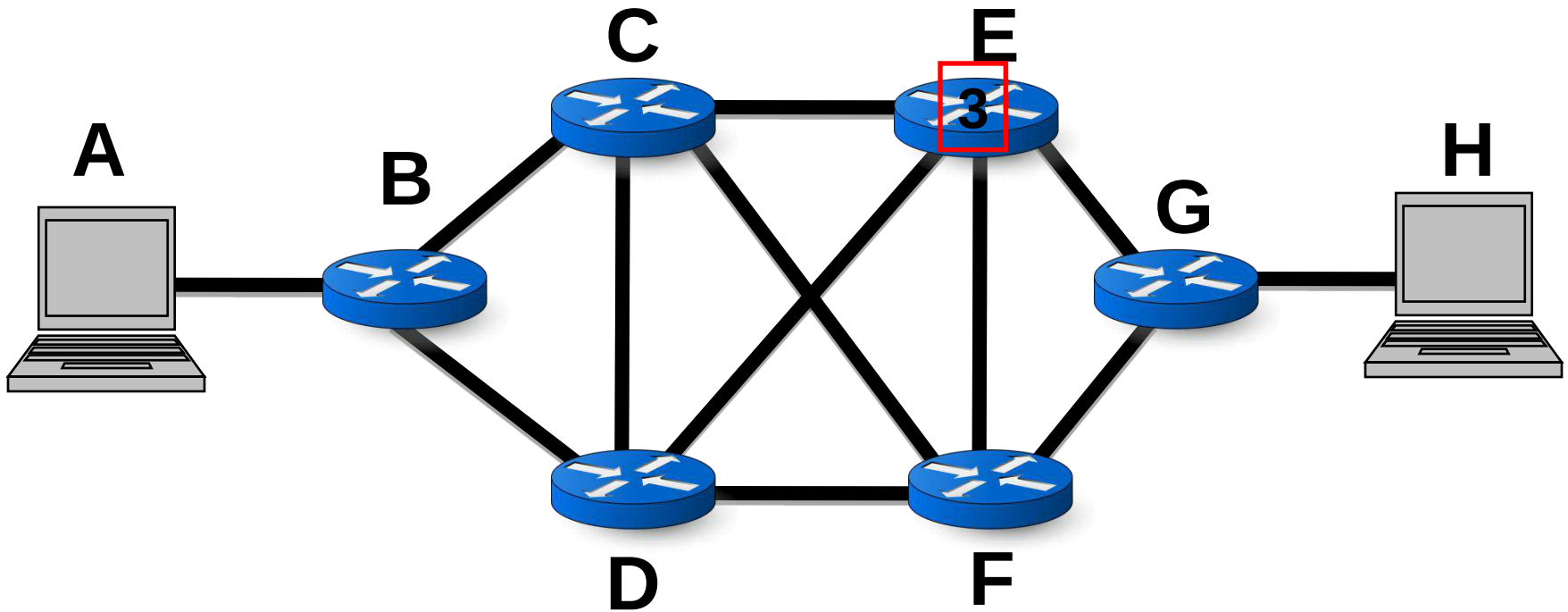
- Packets arrive randomly and chose different paths
- A sequencing number is added to each packet
- This helps in re-ordering the packets at destination and reconstruct the original message

PACKET SWITCHING



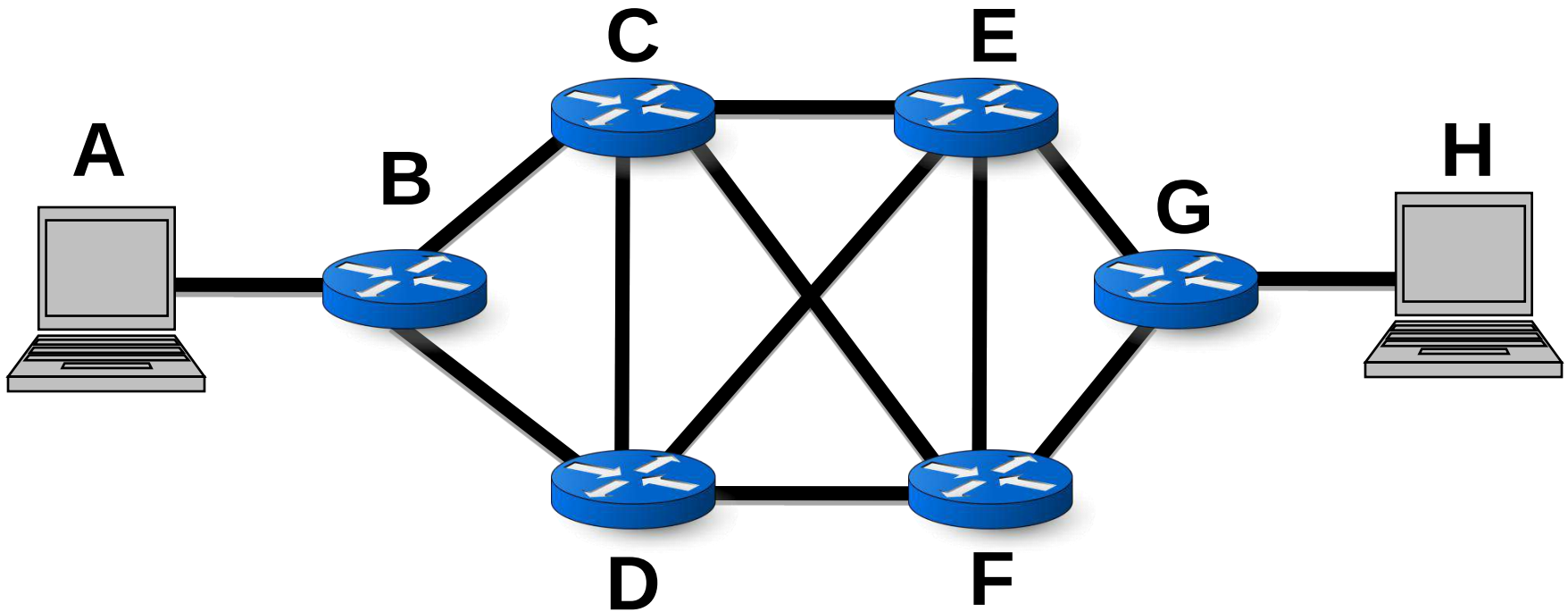
5	4	3	2	1
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PACKET SWITCHING



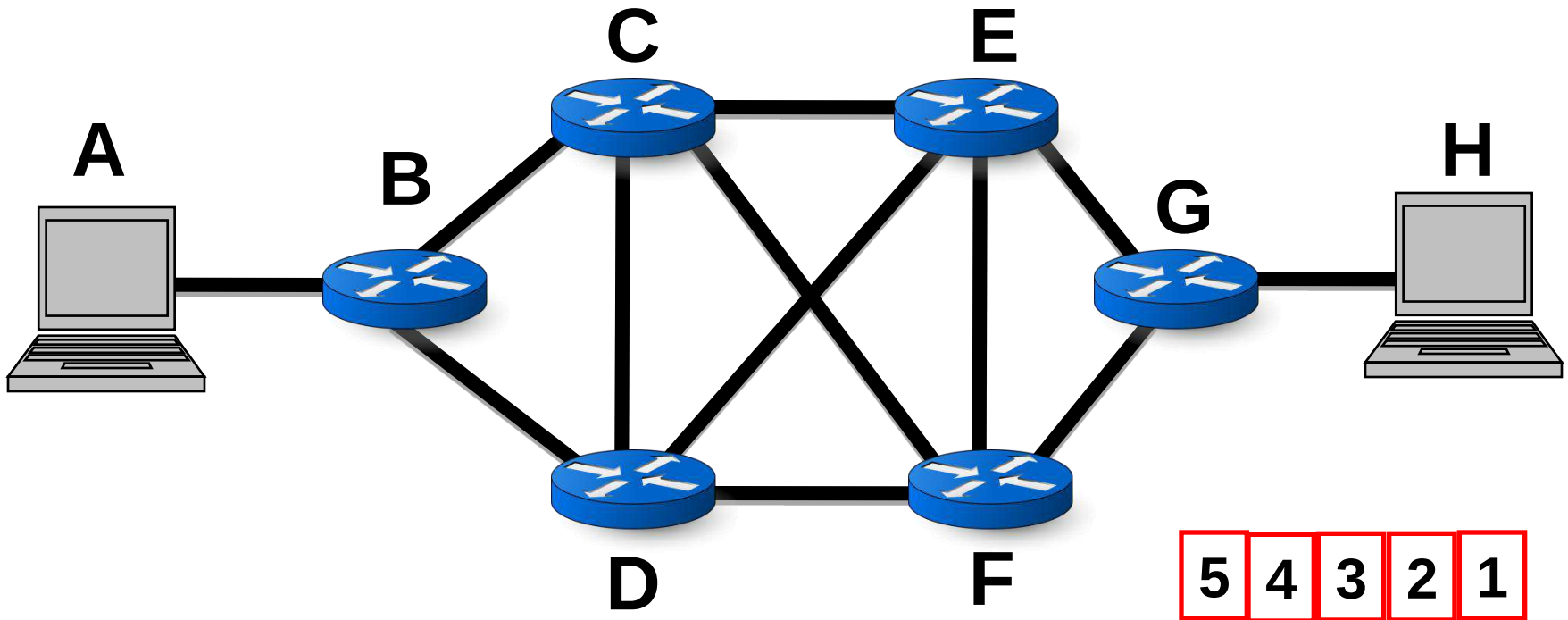
1 2 4 5

PACKET SWITCHING



1 2 4 5 3

PACKET SWITCHING



DATAGRAM PACKET **SWITCHING - BENEFITS**

- Cost effective as storage requirements are minimum and inexpensive
- Less transmission delay
- Packets can be routed around problem links
- Optimal use of communication link (or bandwidth)

DATAGRAM PACKET

SWITCHING -

CONSIDERATIONS

- High implementation costs as complex protocols are used
- Loss of packet
- Retransmission required

VIRTUAL CIRCUIT PACKET SWITCHING

- Virtual Circuit communication is a combination of **circuit switching** & **packet switching**
- The term '**virtual**' signifies that the channel **appears** to be dedicated to the user for a conversation

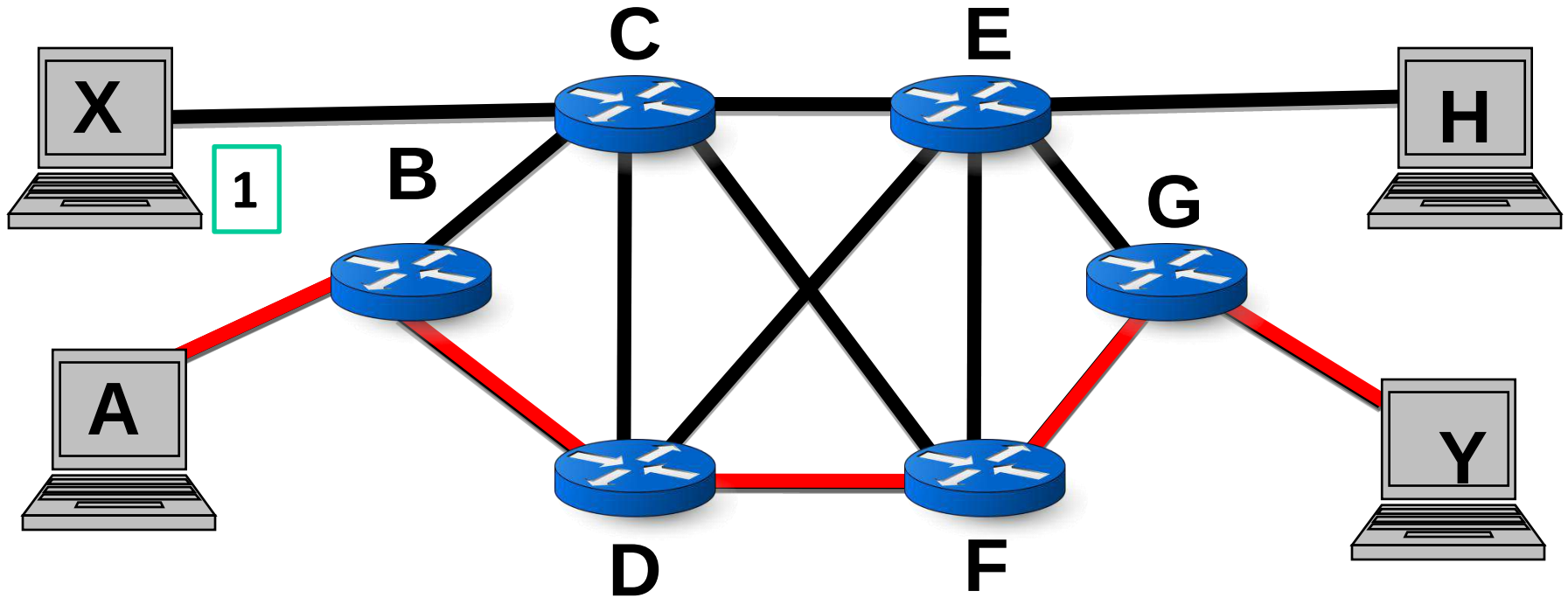
VIRTUAL CIRCUIT **PACKET SWITCHING**

- However; in actual, the devices over the dedicated channel can be shared/used by other devices also for packet forwarding
- Priority may be associated with the packets belonging to the virtual circuit communication
- Examples of virtual circuit switching are **X.25** and **Frame Relay**

VIRTUAL CIRCUIT PACKET SWITCHING

- Circuit switching provides constant bit rate and latency, while these may vary in a **virtual circuit** service because of reasons such as:
 - varying packet queue lengths
 - varying bit rate
 - varying load

VIRTUAL CIRCUIT SWITCHING



5	4	3	2	1
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VIRTUAL CIRCUIT

PACKET SWITCHING -

BENEFITS

- Packets are delivered in order since they all take the same route
- The overhead in the packets is smaller, since there is no need for each packet to contain the full address

VIRTUAL CIRCUIT

PACKET SWITCHING -

BENEFITS

- The connection is more reliable, network resources are allocated at call setup so that even during times of congestion, provided that a call has been setup, the subsequent packets should get through
- Billing is easier, since billing records need only be generated per call and not per packet.

VIRTUAL CIRCUIT

PACKET SWITCHING -

CONSIDERATIONS

- The switching equipment needs to be more powerful, since each switch needs to store details of all the calls that are passing through it and to allocate capacity for any traffic that each call could generate

VIRTUAL CIRCUIT

PACKET SWITCHING -

CONSIDERATIONS

- Resilience (Flexibility) to the loss of a trunk is more difficult, since if there is a failure all the calls must be dynamically re-established over a different route.

NETWORK SERVICE MODEL

**All services provided on the internet
are broadly classified**

RELIABLE

DATAGRAM (non-reliable)
/ Best Effort)

CONNECTION
ORIENTED

CONNECTION LESS

LAYER-WISE **SWITCHING**

SWITCHING TYPE	LAYER WISE IMPLEMENTED ON
Circuit Switching	Physical Layer
Virtual Circuit Switching	Data Link Layer
Datagram Switching	Network Layer

NETWORK LAYER FUNCTIONS

- Host addressing
- Connection model
- **Message forwarding**

ROUTING SCHEMES

ROUTING SCHEMES

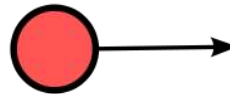
UNICAST

BROADCAST

MULTICAST

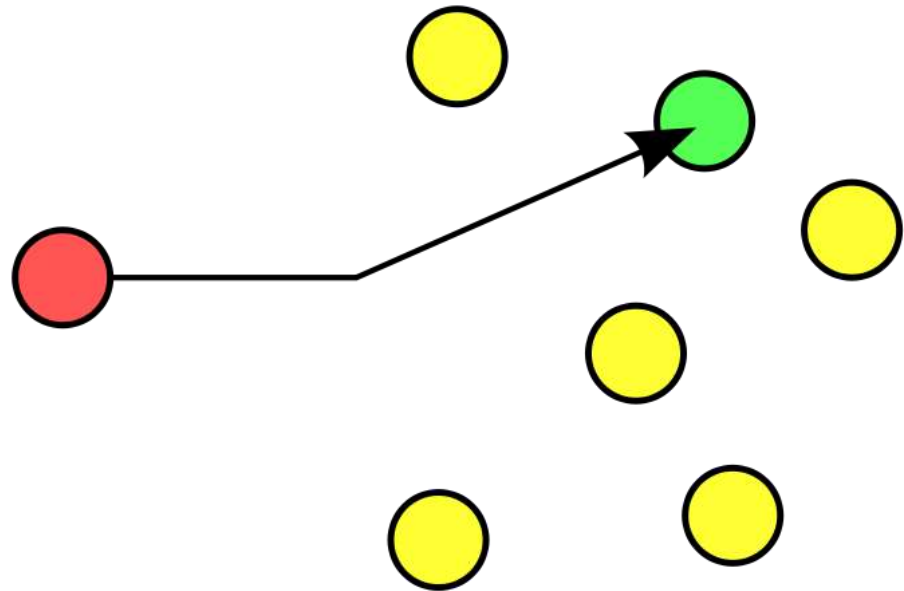
ANYCAST

GEOCAST



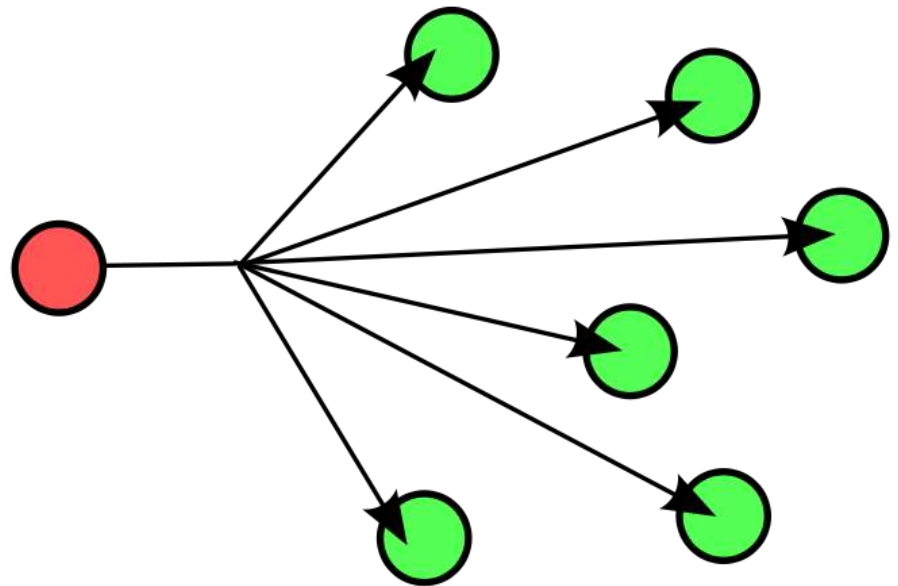
ROUTING SCHEMES

- **UNICAST** - delivers a message to a single specific node



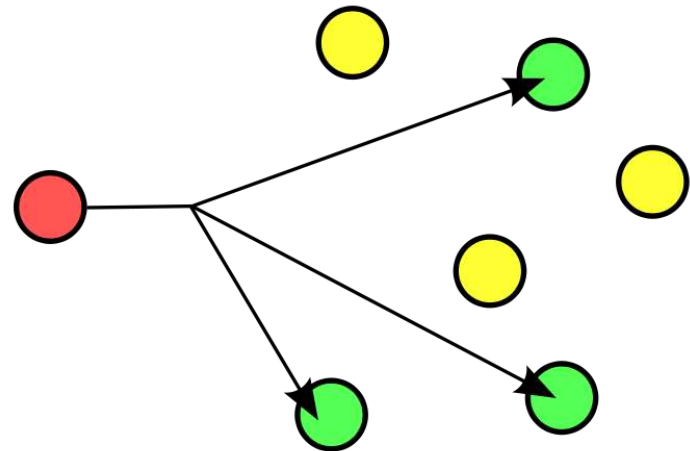
ROUTING SCHEMES

- **BROADCAST** - delivers a message to all nodes in the network



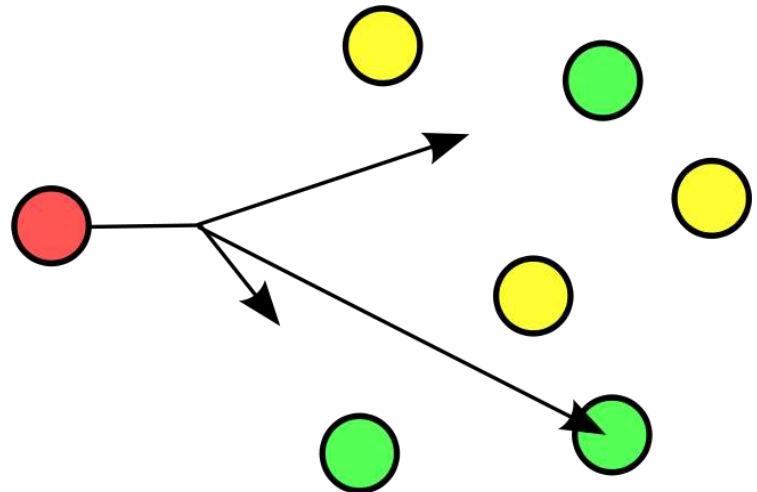
ROUTING SCHEMES

- **MULTICAST** - delivers a message to a group of nodes that have expressed interest in receiving the message



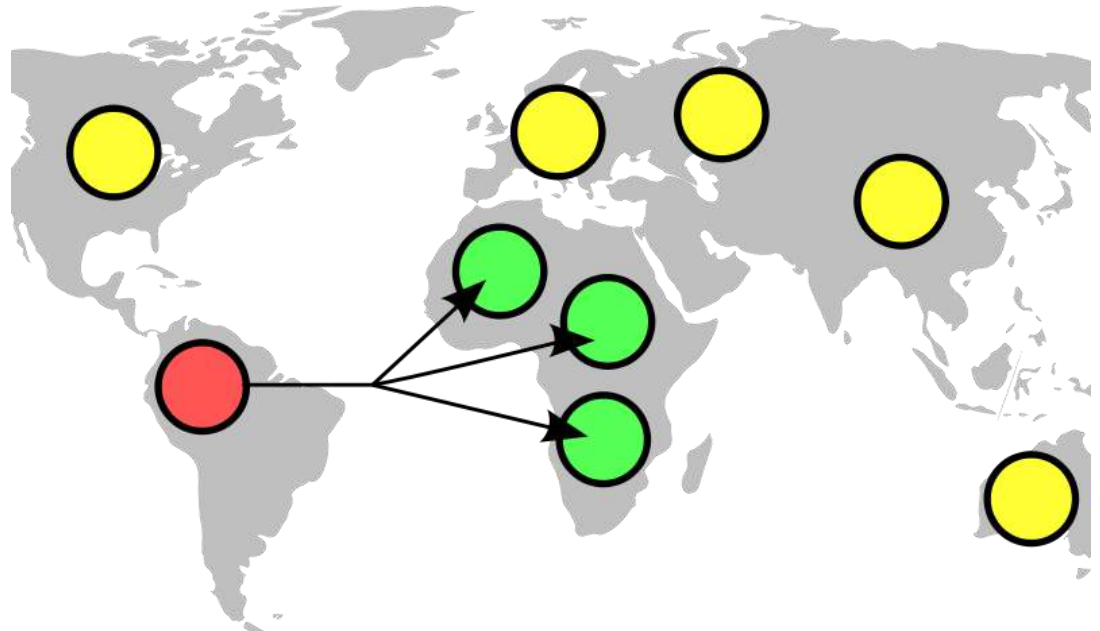
ROUTING SCHEMES

- **ANYCAST** - delivers a message to any one out of a group of nodes, typically the one nearest to the source



ROUTING SCHEMES

- **GEOCAST** - delivers a message to a geographic area



WHAT IS ROUTING

- **Routing** is the task of selecting paths in a network along which to send **network traffic** or **information**
- Routing enables us to connect two or more networks using different network addresses i.e. Internetworking

WHAT IS ROUTER

- A computer or a device that specialize in sending packets over the data network
- Responsible for interconnecting networks
- Forwards packets to their destination by selecting the best path for a packet to travel

ROUTING TABLES

- The routing process usually directs forwarding on the basis of ***routing tables*** which maintain a record of the routes to various network destinations

ROUTING ACTIVITIES

- Routing involves the following two activities:
 - Path determination
 - Packet forwarding

PATH DETERMINATION

- The process of obtaining path in routing table is called **path determination**
- Path selection involves applying a routing metric to multiple routes, in order to select (or predict) the best route

PATH DETERMINATION

- There are three different methods to which router can learn path.
 - Automatic detection of directly connected network.
 - Static & Default routing
 - Dynamic routing

PACKET FORWARDING

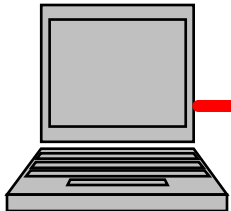
- If a path is established using the information from the **routing table/s**, then the concerned router sends the packet to the next device on the selected path
- The router will perform packet forwarding only if route is available in the routing table

ROUTERS AND OSI MODEL

- **Routing** is; in general, associated with the Third Layer in OSI Model i.e. **Network Layer**
- **Routers**; however are designed to have Network as well as Data Link and Physical Layer Protocols also

ROUTERS AND OSI MODEL

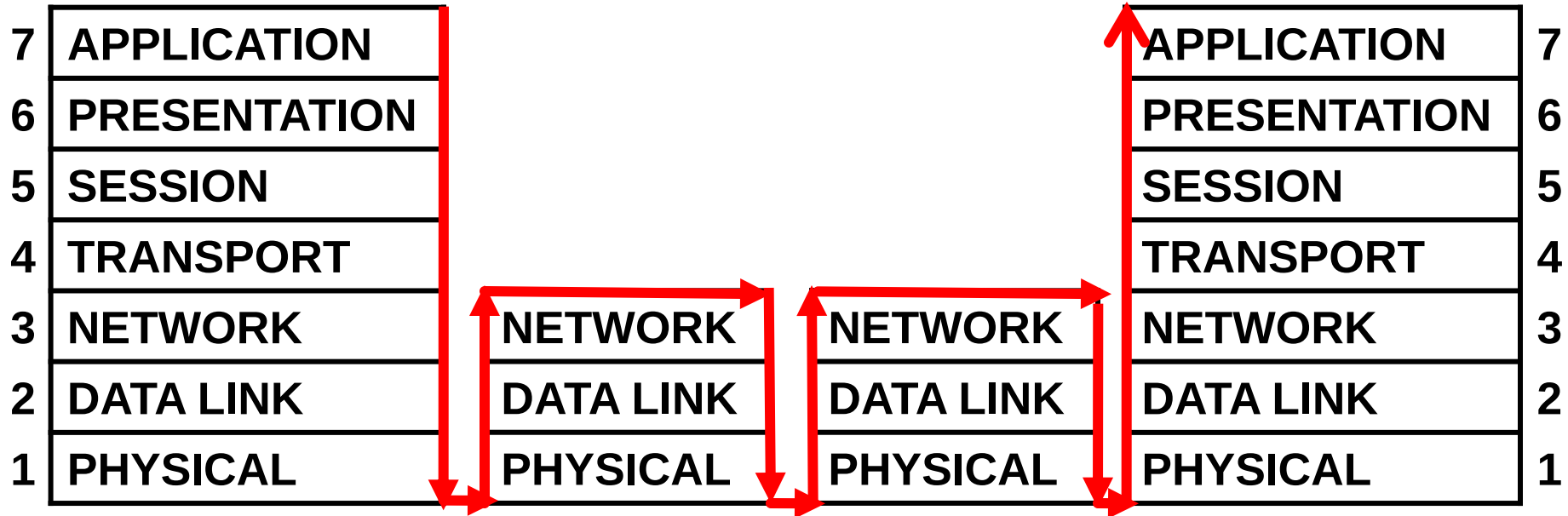
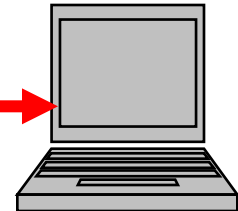
PC 1



Routers Operate at Layer 1,2 & 3



PC 2



ROUTERS AND OSI **MODEL**

- Routers use destination IP address to forward packets
- Router determines the best path using the routing table
- Packet is encapsulated into a frame
- Frame is placed on network medium in form of Bits

ROUTING ALGORITHM

- A router with the help of certain algorithms calculates the **best path** for the packet to reach the destination
- These algorithms are called **routing algorithms**
- This is the way in which a router finds the correct entry in its ***routing table*** and updates the same

ATTRIBUTES OF A ROUTING ALGORITHM

- **Optimality**
 - capability of the routing algorithm to select the best route
- **Simplicity and low overhead**
 - must offer its functionality efficiently, with a minimum of software and utilization overhead
- **Robustness and stability**
 - should perform correctly in the face of unusual or unforeseen circumstances

ATTRIBUTES OF A ROUTING ALGORITHM

- **Rapid convergence**
 - Convergence is the process of agreement, by all routers, on optimal routes
- **Flexibility**
 - should quickly and accurately adapt to a variety of network circumstances

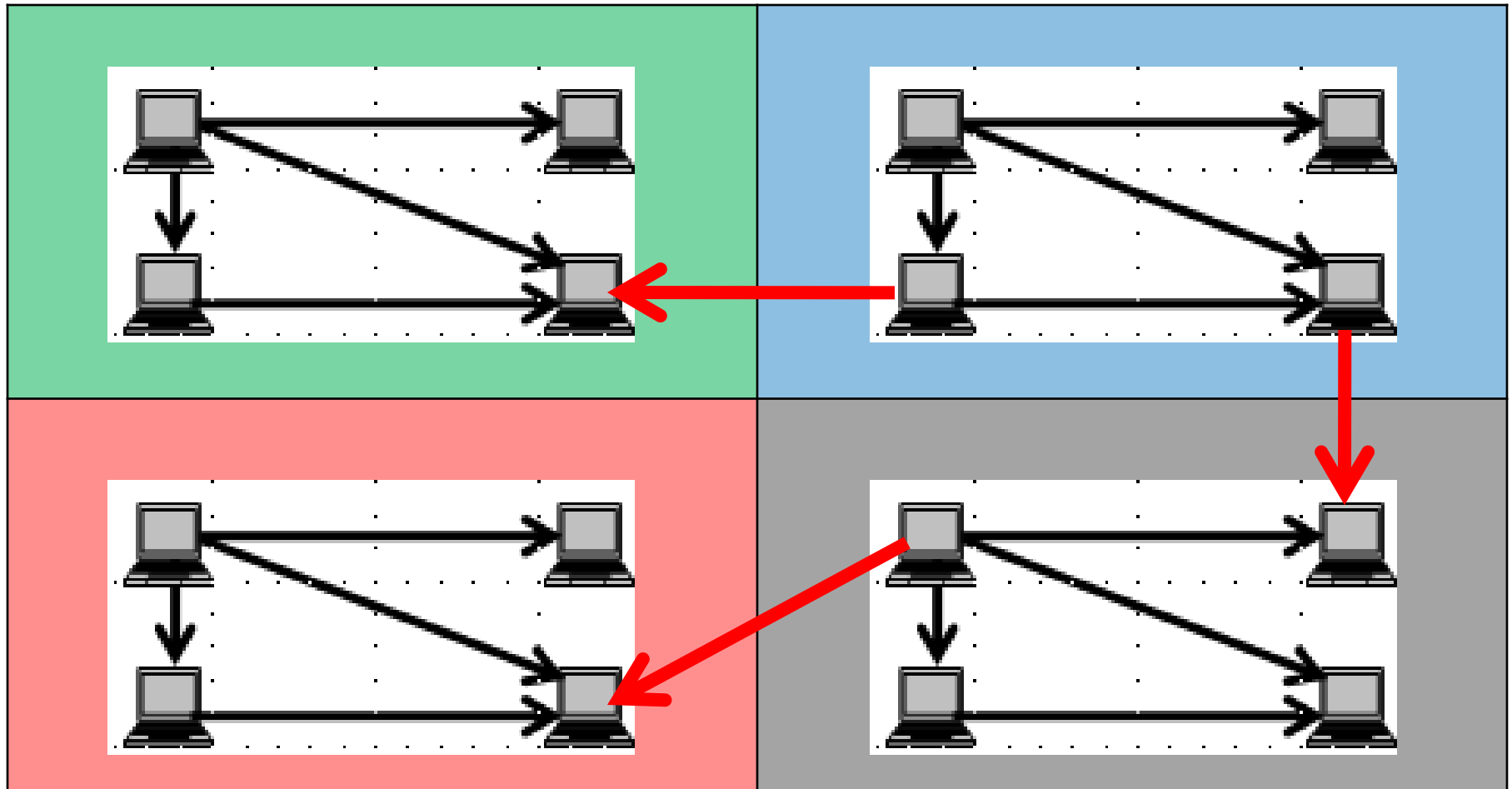
ROUTING METRICS

- Path length
- Reliability
- Delay
- Bandwidth
- Load
- Communication cost

AUTONOMOUS SYSTEMS

- Internet is divided into **autonomous systems**
- An **autonomous system** is a group of network and routers under the authority of a single administration

AUTONOMOUS SYSTEMS



ROUTING PROTOCOLS

- Routing inside an autonomous system is referred to as **intra-domain** routing. For example:

RIP : Routing Information Protocol

- Implementation of **Distance Vector Protocol**

OSPF : Open Shortest Path First

- Implementation of **Link State Protocol**

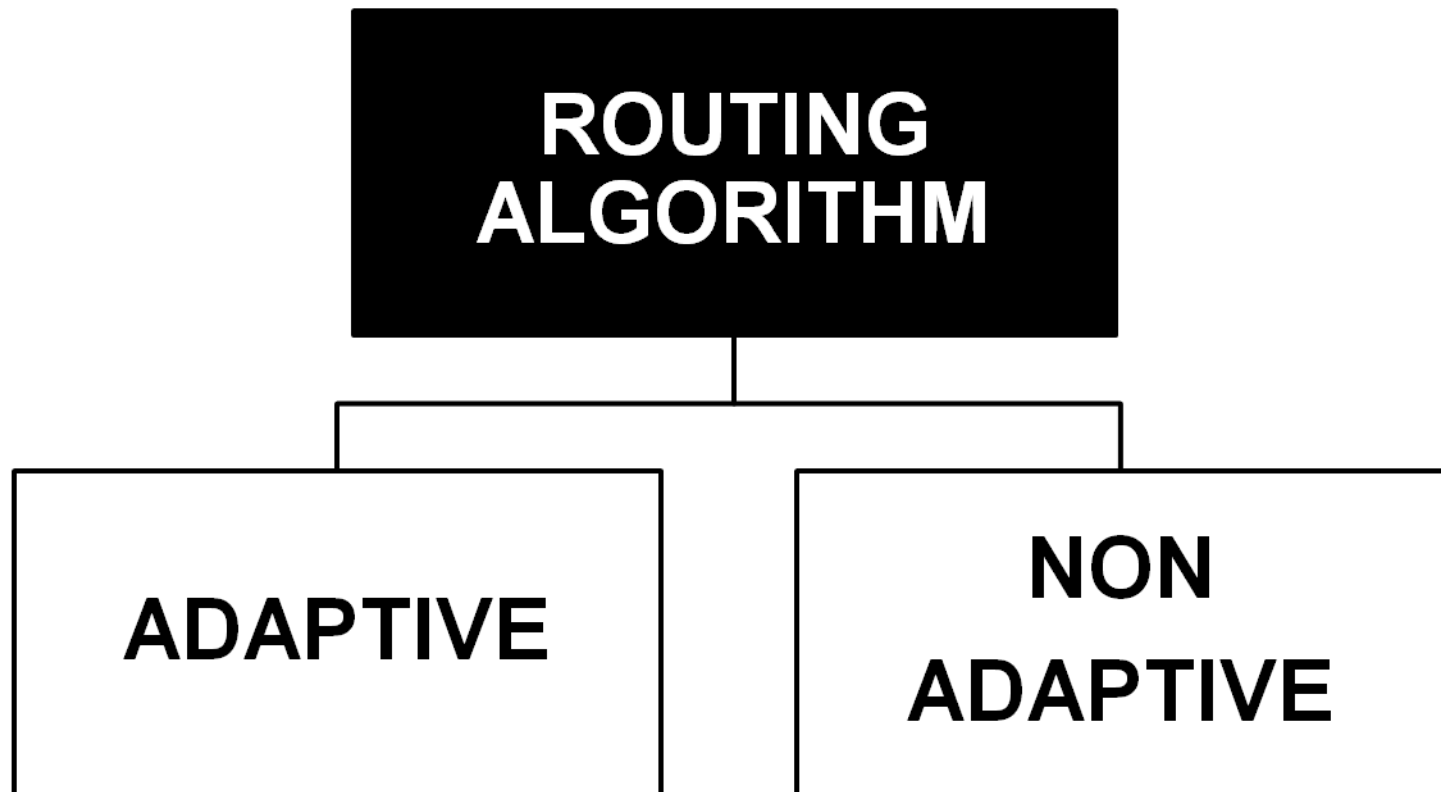
ROUTING PROTOCOLS

- Routing between two autonomous systems is referred to as **inter-domain** routing. For example:

BGP : Border Gateway Protocol

- Implementation of Path Vector Protocol

ROUTING ALGORITHM



ROUTING ALGORITHM

- SINGLE-PATH **versus** MULTIPATH
- FLAT **versus** HIERARCHICAL
- HOST-INTELLIGENT **versus** ROUTER-INTELLIGENT
- INTRADOMAIN **versus** INTERDOMAIN
- LINK-STATE **versus** DISTANCE VECTOR

IS PC A ROUTER ?

- PCs can also forward packets and perform routing, though they are not specialized hardware and may suffer from limited performance

SUGGESTED READING

- BOOKS:
 - Andrew S Tanenbaum, Computer Networks, Pearson Education
 - Behrouz A. Forouzan, Data Communications and Networking, Tata McGraw
- ONLINE RESOURCES
 - docwiki.cisco.com
 - en.wikipedia.org

TOPICS

- ❖ NETWORK LAYER - BASICS

- ❖ NETWORK LAYER - FUNCTIONS

 - ☐ CONNECTION MODEL

 - ☐ ADDRESSING

 - ☐ ROUTING

Thank You